



SIMATS Online Programming Lab

DS PROGRAMMING



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QUESTIONS

10 / 10 completed

Doubly Linked List

DOUBLY LINKED LIST

"A music player maintains a playlist where users can move both forward and backward between songs. This can be efficiently implemented using a doubly linked list. Write a C program to create a doubly linked list and display nodes in forward and backward directions.

Test Case 1

Input

10 → 20 → 30 → 40

Output

Forward Traversal : 10 20 30 40

Backward Traversal: 40 30 20 10

"

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 typedef struct Node {
5     int data;
6     struct Node *prev;
7     struct Node *next;
8 } Node;
9
10 Node* createNode(int value) {
11     Node* newNode = (Node*)malloc(sizeof(Node));
12     if (newNode == NULL) {
13         printf("Memory allocation failed!\n");
14         exit(1);
15     }
16     newNode->data = value;
17     newNode->prev = NULL;
18     newNode->next = NULL;
19     return newNode;
20 }
21
```

OUTPUT

```
Enter number of nodes: Enter values: Forward Traversal : 20
30 40 40 40 40 40 40 40 40
Backward Traversal: 40 40 40 40 40 40 40 40 30 20
```

INPUT

10 20 30 40

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